

Revised Generation Interconnection System Impact Study Report

For Queue Project Q16-0623

La Canada 138 kV
January 21, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0623. The Interconnection Customer proposes a 92.7 MW solar photovoltaic facility interconnecting at the La Canada 138 kV substation in the Pennsylvania area. The request was received on May 28, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 242 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,941,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Neevin - Safe Harbor 138.0 kV	Ipl Fit Monrovia I Llc - Safe Harbor 345.0 kV L	119.0	227	7.2
Irvine Substation - Marcus Hook Energy Lp 500.0 kV Line	Marcus Hook Energy Lp - Guilbeau Electric 500.0 kV	101.1	2,715	6.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Neevin - Safe Harbor 138.0 kV	Rebuild line with higher-capacity conductor	\$6,909,000	70% (\$4,860,000)
Irvine - Marcus Hook Energy Lp 500.0 kV	Reconductor line	\$7,513,000	1% (\$81,000)
	Total allocated to project		\$4,941,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.